

Evaluation of Home-Based Follow-Up of High-Risk Elderly Patients Discharged from Hospital

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ABSTRACT

Objective: To evaluate pharmacist-conducted follow-up at home of high-risk elderly patients discharged from hospital.

Method: A randomised controlled study, in which medical patients admitted to hospital and fulfilling high-risk criteria (including age 60 years or older and prescribed four or more regular medications) were randomly assigned to an intervention or control group. Patients in the intervention group were visited at home by a pharmacist five days after discharge. The pharmacist educated patients on their medications, encouraged compliance, assessed for drug-related problems, intervened when appropriate and communicated all relevant findings to community health professionals. The intervention group patients were revisited at home 90 days after discharge to evaluate the outcomes of interventions made on day 5. Patients in the control group were visited at home by a pharmacist 90 days after discharge and provided with an identical comprehensive medication review.

Results: One hundred and twenty-one patients completed the study. There were no significant differences between the two groups in key clinical and demographic parameters at baseline. A median of three drug-related problems were identified in each intervention group patient at the five day home visit. Ninety days after discharge this had declined to one, compared to two for the control group patients ($p < 0.0001$). In the intervention group, compliance had improved and was significantly higher than for the control group after 90 days ($p < 0.0001$). There was a significant decline in the use of nonsteroidal anti-inflammatory drugs by the intervention group patients. Forty-five per cent of the control group patients had unplanned readmissions to hospital during the 90 days following discharge compared to 28% of the intervention group patients ($p = 0.05$). The intervention program was well-received by patients and their general practitioners. Recommendations from the pharmacist were implemented by general practitioners in 79% of cases.

Conclusion: A pharmacist-conducted follow-up at home of high-risk elderly patients discharged from hospital is valuable in identifying and addressing drug-related problems and reducing the risk of readmission to hospital.

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INTRODUCTION

Problems associated with the use of medication in society continue to be a significant public health burden. The Second National Report on Patient Safety 'Improving

Medication Safety' of the Australian Council for Safety and Quality in Health Care concluded: 'Between two and three per cent of hospital admissions are related to problems with medicines (approximately 140 000 per year). This is more than the combined number of admissions for asthma and heart failure. Of the 100 million encounters recorded in general practice each year, it is estimated that around 400 000 of these are related to adverse medication events.'¹

Approximately 12% of all admissions to hospital medical wards in Australia,²⁻⁴ including at the Royal Hobart Hospital,⁴ are drug-related. About half of these hospital admissions are considered to be avoidable.¹⁻³

Elderly patients discharged from hospital are at a high risk of experiencing adverse drug events.^{5,6} Alterations to a patient's medication regimen (drugs or dosages) may occur during hospitalisation and medications may not be taken appropriately after discharge. Documented errors include continuation of medications that were discontinued at discharge, failure to start new medications, taking the incorrect dosage, and a lack of understanding of directions for medication use.^{5,6}

Unplanned readmissions to hospital have a substantial impact on healthcare costs.^{7,8} Problems with medication management have previously been identified as a major cause of readmission to hospital.^{7,9-13} The majority of drug-related problems (DRPs) identified in studies of unplanned readmissions to hospital are potentially preventable and the types of problems found have indicated that interventions should be system-based and directed at both health professionals and patients.^{10,12}

The objective of this study was to undertake a randomised controlled evaluation of a pharmacist-conducted follow-up at home of high-risk elderly patients discharged from hospital.

METHOD

The study was approved by the Royal Hobart Hospital Research and Ethics Committees. Project information was sent to the local Division of General Practice and local branch of the Royal Australian College of General Practitioners, prior to the commencement of the study.

Patient Selection

Patients were recruited from the Royal Hobart Hospital (the only major public hospital in the southern region of Tasmania) a 400-bed acute care teaching hospital. Patients admitted to the medical units at the hospital between November 2000 and December 2001 were eligible to participate in the study, if they were aged 60 years or older, had at least two chronic medical conditions requiring medication (including at least one of heart failure, ischaemic heart disease, chronic obstructive airways disease or diabetes mellitus) and were prescribed

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four or more regular medications. Patients were excluded if they lived in a domiciliary care facility or beyond the greater Hobart area, were to be visited at home by a community nurse within five days of discharge, had terminal malignancy or were unable to provide informed consent.

Patients were allocated to either an intervention or control group by the study pharmacist (MN) responsible for conducting the home visits, using a computer-generated list of random numbers. The study pharmacist was not an employee of the Royal Hobart Hospital. Randomisation occurred the day after patients were discharged from hospital. A review of medical records and initial interview was conducted with all patients in hospital to document baseline characteristics. Baseline data collected for each patient included demographic features, reason for admission, smoking habits, alcohol intake, medical history, medication history and management during hospitalisation.

Home Visit Protocol

Five days after discharge from hospital, the intervention group patients were visited at home by the study pharmacist. Patients were telephoned one day prior to the scheduled home visit to arrange a suitable time for the review. The objectives of this visit were to educate patients about their medications, answer any queries from patients or their caregivers, optimise medication management and improve compliance, detect DRPs and improve liaison with community-based health services.

The study pharmacist performed a pill count to assess compliance with the medication regimen. Prior to discharge, medications, dosages and quantities supplied were noted. The pharmacy department at the Royal Hobart Hospital provides five days of medication to each patient at discharge. Patients were deemed noncompliant if they had taken more or less medication than prescribed. Also during the visit, patients were asked a modified compliance question:¹⁴ *'People often have difficulty taking their pills for one reason or another ... have you had any difficulty taking your pills—about how often would you say you miss taking your pills?'* Response options consisted of 'once a day', 'more than once a week, but less than once a day', 'once a week', 'once a month', 'rarely', or 'never'.

Patients who were deemed noncompliant (on pill count) or who appeared to have poor understanding of their medications were either offered a compliance device (e.g. dosette) or their community pharmacist was requested to provide additional services such as filling and delivering a compliance device (e.g. Webster pack). Where possible, caregivers were requested to provide increased support or community nurses asked to provide services such as filling a compliance device.

The study pharmacist performed a comprehensive medication review to assess the need for all medications and identify any drug-related issues. The clinical pharmacy service complied with The Society of Hospital Pharmacists of Australia's Standards of Practice for Clinical Pharmacy.¹⁵ DRPs were pre-defined utilising a modified categorisation by Cipolle et al.¹⁶ and *Drug Interactions Facts*¹⁷ was used to categorise drug interactions. Only interactions reaching a significance of level 1 (potentially severe or life-threatening interaction; occurrence has been suspected, established or probable in well-controlled studies) or level 2 (interaction may cause deterioration in a patient's clinical status; occurrence suspected, established or

probable in well-controlled studies) were recorded as being clinically relevant.

A brief letter outlining the patient's medication regimen and any suggested changes or monitoring procedures, was composed in the patient's home and given to them to present to their doctor. Soon after the home visit, the study pharmacist telephoned the patient's general practitioner and community pharmacist to inform them of the study and to discuss any urgent issues.

The intervention group patients were revisited at home 90 days after discharge to evaluate the outcomes of interventions made on day 5.

There was no contact with the control group patients (after the initial collection of baseline characteristics which occurred during hospitalisation) until 90 days after discharge from hospital. They were then visited at home and provided with a comprehensive medication review in an identical manner to the intervention group five days following discharge, including the subjective assessment of compliance. The patient's general practitioner and community pharmacist were contacted to verify the patient's medication regimen.

Outcomes

The primary endpoints for this study were the number of unplanned readmissions to hospital within 90 days of initial discharge, total days of readmission and out-of-hospital deaths (ascertained, when necessary, by contacting relatives or the patient's general practitioner and reviewing hospital records). Medical records were accessed to monitor for emergency department visits or hospital admissions. Patients were also asked during the 90-day visit if they had been admitted to any other hospitals. Drug-related readmissions were assessed by reviewing documentation in the medical notes. Patients were deemed to have a drug-related readmission if it was documented by the medical team treating the patient.

Other outcomes included the number of drug-related issues identified, number of prescribed medications, level of compliance and number of patients taking nonsteroidal anti-inflammatory drugs (NSAIDs), alone and in combination with either angiotensin converting enzyme (ACE) or angiotensin II (AII) inhibitors. A multiple-choice 'satisfaction questionnaire', to be completed anonymously, was given to each intervention group patient after the 90-day follow-up, along with a reply-paid envelope.

Statistical Analysis

Statistical analyses were performed using Statview 5.01 (Abacus Concepts Inc., Berkeley, CA, USA). Non-parametric techniques were principally used to describe patient characteristics and examine differences in variables between the groups of patients (Mann-Whitney U test) and within groups between study periods (Wilcoxon signed ranks test). Categorical data were analysed by Chi-square or Fisher's exact tests; the latter was only used when at least one of the variables had less than five patients or events. A p-value less than 0.05 was considered statistically significant.

RESULTS

A total of 402 patients initially met the study's inclusion criteria, of whom 266 were excluded for various reasons (Table 1). Fifteen patients (8 intervention, 7 control) had

to be excluded after randomisation. Two control group patients were excluded because they died before discharge. One control group patient was involved in a fatal car accident shortly after discharge. Two patients (1 intervention, 1 control) voluntarily withdrew from the study. Five patients (3 intervention, 2 control) were unable to be contacted at follow-up. Three patients (2 intervention, 1 control) who were discharged home were later admitted to a nursing home. Two intervention group patients were excluded as they were discharged with intensive nursing follow-up. The excluded patients were not included in the baseline analyses.

Table 1. Patient exclusion from study prior to randomisation

Reason for exclusion	Number (n = 266)
Nursing home resident	87
Living > 50 km from Royal Hobart Hospital	71
Seen by community nurse within 5 days of discharge	55
Dementia	18
Terminal cancer	14
Refused to participate	12
Required interpreter	9

Table 2 summarises the clinical and demographic features of the 121 patients who completed the study. There were no significant differences between the two groups in any of the variables.

There were 174 drug-related issues or problems identified in the intervention group five days after discharge and this was reduced to 45 issues by the 90-day follow-up. In comparison, there were 152 drug-related issues identified in the control group at the 90-day follow-up. A median of three DRPs per patient was identified in the intervention group at the five-day visit. After 90 days, this had declined to one (Wilcoxon signed ranks test, $z = -5.8$, $p < 0.0001$), compared to two for the control group patients (Mann-Whitney $U = 634.5$, $z = -5.4$, $p < 0.0001$). Sixty-five per cent of the intervention group patients, compared to 86% of the control group patients, had at least one DRP at the 90-day follow-up (chi-square = 7.3, $df = 1$, $p < 0.01$).

Drug-related issues are summarised in Table 3. Thirty-eight per cent of the intervention group patients were found to be noncompliant (using a pill count), with one or more of their medications five days after discharge. These patients received remedial counselling and were offered a compliance device/reminder routine. Nine patients (16%) were provided with a dosette or arrangements made for a Webster pack to be filled and delivered to their home. Three patients were given a spacer to enhance compliance with their inhalers. Five patients were provided with accessories for their nebuliser as it was deemed that they were not receiving adequate delivery of the nebulised drug. Two patients had their nebuliser serviced. Four patients were found to be not coping very well so arrangements were made for community nurses to assist with their medication.

The interview and pill-count measures of compliance were compared in the intervention group patients at the five-day visit. The sensitivity of the interview question (i.e. the subjective measure of compliance) was relatively

high (86%), and the overall accuracy on the interview method was approximately 67%. However, the positive predictive value was relatively low (55%). The intervention group had improved compliance at the 90-day follow-up and there was a significant difference between the intervention and control groups (Table 4). When the intervention group patients were asked how often they forgot to take their medications, 39% answered that they 'never' forgot their medications at the five-day follow-up visit. After 90 days this had increased to 87% whereas the control group reported that they 'never' forgot to take their medications in 44% of cases (chi-square = 22.8, $df = 1$, $p < 0.001$). Compliance was not significantly related to the number of prescribed regular medications at either 5 or 90 days after discharge.

The use of NSAIDs was relatively common in these elderly patients. Most patients taking an NSAID were also receiving an ACE or AII inhibitor; in fact, more than 20% of all patients were using this combination on admission to hospital. Approximately half of the patients in the intervention group and the control group had their NSAID ceased prior to discharge. However, the majority of intervention group patients who had had the NSAID stopped during their admission were taking it again when visited after five days. After consultation with the patient, their general practitioner and community pharmacist, the majority of these patients were re-instructed to cease the

Table 2. Baseline characteristics

Patient characteristics	Intervention (n = 57)	Control (n = 64)
Gender (male)	44%	31%
Age (median and range)	74 (65-90)	77 (60-91)
Living alone	47%	45%
English as second language	9.0%	7.0%
Primary admission diagnosis		
Cardiac disease	56%	54%
Respiratory disease	14%	11%
Gastrointestinal disease	4.0%	9.0%
Other	26%	26%
Nil or moderate alcohol intake	92%	91%
Smoker or ex-smoker	67%	63%
Seen by hospital pharmacist at discharge	89%	89%
Nursing assistance at home	28%	21%
Hospital stay, in days (median and range)	7 (2-42)	6 (1-94)
Unplanned admissions in previous 12 months (median and range)	0 (0-11)	0 (0-4)
Unplanned readmission within the previous 12 months	47%	42%
Chronic medical conditions (median and range)	5 (2-9)	5 (2-13)
Regular medications on admission (median and range)	7 (2-15)	6.5 (1-16)
Regular medications on discharge (median and range)	8 (3-15)	8 (3-16)

Table 3. Drug-related issues

Issue	Intervention 5 days (% patients)	Intervention 90 days (% patients)	Control 90 days (% patients)	P-value*
Drug interaction	46	43	50	
Non-compliance†	38	5.6	22	< 0.01
Drug not given or taken when indicated	30	11	27	< 0.05
Difficulty getting prescription	21	1.9	0	
Drug given or taken without indication	18	5.6	6.9	
Using drug delivery system incorrectly	18	1.9	26	< 0.001
Excessive or insufficient dosage	16	5.6	28	< 0.01
Drug given at incorrect time	16	0	25	< 0.001
Drug used not safest or most efficacious	16	0	16	< 0.01
Dispensing error	11	1.9	5.1	
Taking discontinued medications	11	0	5.1	
Drug allergy/adverse reaction	8.8	3.8	3.4	
Hoarding medications	7.0	0	0	
Patient not responding to therapy	5.3	0	5.1	
Unable to open medications	3.5	1.9	1.7	
Drug-food interaction	1.8	0	1.7	
Improper route of administration	1.8	0	0	
Therapeutic contraindication	1.8	0	5.2	
Duplicating medications	1.8	0	1.7	
Drug level monitoring required	0	1.9	8.6	
Other	32	1.9	21	< 0.01

*For significant differences observed between the intervention and control groups at 90 days.

†Does not include patients with other categories of drug-related issues (e.g. difficulty getting prescription).

Table 4. Summary of principal outcome measures

	Intervention 90 days (n = 57)	Control 90 days (n = 64)	P-value
Number of patients with unplanned readmissions	16 (28%)	29 (45%)	0.05
Total days as hospital readmissions (median and range)	0 (0-30)	0 (0-50)	0.06
Deaths (%)	3 (5%)	5 (8%)	
Number of regular medications (median and range)	8 (3-16)	8 (3-17)	
Number of patients with at least one drug-related issue*	35 (65%)	51 (86%)	< 0.01
Drug-related issues identified per patient (median and range)	1 (0-5)	2 (0-8)	< 0.001
Number of patients with self-reported compliance ('never miss medication')*	47 (87%)	26 (44%)	< 0.001
Number of patients taking NSAID*	5 (9%)	15 (24%)	< 0.05
Number of patients taking NSAID + ACE/AII inhibitor*	5 (9%)	12 (20%)	

*Percentage of patients alive at 90 days in each group (54 intervention, 59 control).

NSAID = nonsteroidal anti-inflammatory drug; ACE = angiotensin converting enzyme; AII = angiotensin II

NSAID; this cessation was sustained, as only 9% of the patients were taking the combination after 90 days. The control group also appeared to frequently restart their NSAID. Significantly more patients in the control group were taking NSAIDs than in the intervention group 90 days after discharge (24% vs 9%; chi-square = 4.2, df = 1, p < 0.05).

There were 67 (25 intervention, 42 control) unplanned readmissions during the 90 days after discharge. Forty-

five per cent of patients in the control group were readmitted, compared to 28% in the intervention group (chi-square = 3.8, df = 1, p = 0.05). The length of time spent with the patient at five days was not related to the risk of unplanned readmission in the intervention group. Other factors not associated with readmission were the length of time spent in hospital on the first admission and the number of chronic medications at initial discharge from hospital.

Table 5 displays the reasons why patients in the two groups were readmitted. There was a similar distribution of the admission types between the groups. Brief details of the five drug-related readmissions are:

- severe bradycardia secondary to digoxin, atenolol and flecainide;
- unstable angina secondary to stopping atenolol;
- unstable angina plus exacerbation of heart failure secondary to poor compliance with all medication;
- hypoglycaemia secondary to decreasing oral intake without altering insulin therapy; and
- amiodarone toxicity secondary to failing to reduce a loading dosage regimen after discharge.

Table 5. Reasons for unplanned readmissions

	Intervention	Control
Drug-related	3	2
Related to, or an exacerbation of, existing condition	8	17
New condition (seemingly unrelated to previous reason for admission)	5	10

Drug interactions were relatively common in both groups and their prevalence did not change over the course of the study (Tables 3 and 6). However, almost none of the interactions warranted total avoidance. Examples were:

- digoxin plus frusemide (level 1);
- warfarin plus amiodarone (level 1);
- frusemide plus hydrochlorothiazide (level 2);
- atenolol plus diltiazem (level 2);
- simvastatin plus diltiazem (level 2); and
- simvastatin plus gemfibrozil (level 1).

Table 6. Prevalence of drug interactions with time

Group and time	Drug interactions (level 1 or 2)		Drug interactions (level 1 or 2)	
	Interactions	Patients	Interactions	Patients
Intervention 5 days	42	46%	19	25%
Intervention 90 days	32	43%	15	20%
Control 90 days	48	50%	20	24%

Level 1 = potentially severe or life-threatening interaction; occurrence has been suspected, established or probable in well-controlled studies.

Level 2 = interaction may cause deterioration in a patient's clinical status; occurrence suspected, established or probable in well-controlled studies.

The median duration of home visits for the intervention group at five days was 50 minutes (range: 25–120). The median duration of home visits for the intervention group and control group at 90 days was 30 minutes (range: 15–60) and 45 minutes (range: 20–90), respectively.

There was an overwhelmingly positive response from the returned survey (63% response rate), with 94% of patients stating they were 'very satisfied' with the

amount of contact they had with the pharmacist. Eight-four per cent of patients stated that the information they were given by the pharmacist 'helped a great deal', while the remaining 16% stated that the information had 'helped somewhat'. Ninety-four per cent and 6% of patients rated the quality of the service as 'excellent' and 'good', respectively. The acceptance of suggestions made by the pharmacist following the five-day visit was high (79%).

DISCUSSION

A major contributor to inadvertent polypharmacy and DRPs in the elderly appears to be hospitalisation and the consequent changes in medication (drugs or brands of drugs) at the transition from outpatient to inpatient care and back.^{5,10,18-23} Forster et al.⁵ recently demonstrated in their study of 400 consecutive general medical patients (mean age 57 years) that 19% experienced adverse drug events after hospital and that adverse drug events were the most common (66%). Furthermore, they concluded that many of these events could have been prevented or ameliorated with simple strategies.

This study examined medication management in elderly patients recently discharged from hospital, and found clear evidence of suboptimal use of prescribed medications. Overall, medication management was relatively poor, with many previously undetected and potentially serious problems.

Pharmacist-conducted follow-up of high-risk elderly patients at home after discharge resulted in a significant reduction in DRPs and unplanned readmissions and was welcomed by patients and their general practitioners. There was an improvement in compliance, and less drug-related issues that could adversely affect a patient's health. Therapeutic recommendations made to the general practitioners were generally implemented. This study shows that relatively simple investigation and interventions by a pharmacist working closely with patients and caregivers can improve the efficacy and safety of medication use in the elderly.^{24,25}

It has been noted that an assessment of medication management in the home provides an invaluable opportunity to detect and address problems likely to result in poor health outcomes.¹⁸ It is also well-documented in the literature that many patients have a poorly planned discharge and their general practitioner is not fully informed of their admission.²⁶⁻³²

The Second National Report on Patient Safety concluded: '*Patients can leave hospital with a particular medicine, but may experience a breakdown in communication between their specialist and regular GP leading to inappropriate medicines being used. Patients and their carers may be confused by complex instructions, particularly when taking multiple medications. It is essential for patient care that information about a patient's medicines is communicated to the hospital when the patient is admitted and back to their community health professionals when they are discharged. Poor communication at the time of discharge from hospital or errors in prescribing or transcribing at discharge can contribute to medication incidents.*'¹

Non-adherence to medications is common after discharge.^{6,32} Almost 39% of the intervention group patients had either taken more or omitted some of their

medication within five days of discharge. The subjective assessment and objective measure of compliance were compared at five days. The sensitivity of the interview question was relatively high (86%). Previously published studies have indicated that patient interviews can provide a valid measure of compliance.^{14,33,34}

Interestingly, many intervention group patients had difficulty getting a prescription when visited at five days. It is possible that patients who delayed obtaining prescriptions were at risk of an early readmission because of noncompliance with their therapy. Conversely, over 10% of intervention group patients were found to be taking discontinued medications five days after discharge. This again highlights the valuable role of the pharmacist in identifying potential problems and intervening early before negative outcomes result.

There were a large number of drug interactions identified. However, few interactions were deemed to require consultation with the prescriber because they were not considered clinically relevant unless monitoring was not being performed. For example, it was common for patients with heart failure to be treated with an ACE inhibitor and spironolactone. The use of these two drugs in combination can cause hyperkalaemia, especially in the renally compromised without adequate monitoring. However, this combination is advocated in the contemporary management of heart failure. Often interactions were of a level 1 significance, however the combinations were not contraindicated (e.g. warfarin and amiodarone). Hence, while there were significant drug interactions, most did not require ceasing the drug.

The use of an NSAID and ACE or AII inhibitor was chosen as an outcome measure of the quality use of medicines. Both NSAIDs and ACE or AII inhibitors can lead to renal insufficiency.³⁵⁻³⁸ NSAIDs and ACE inhibitors are the main causes of drug-induced acute renal failure in the elderly.³⁸ This combination is nephrotoxic in the elderly and should be avoided, especially in those taking diuretics.³⁵⁻³⁸ It was of some concern to note that almost every patient who was taking an NSAID was also taking an ACE or AII inhibitor. The intervention reduced the combined use of these drugs, with the decline almost reaching statistical significance. We have also shown that patients recommence medications that have been stopped during their hospital stay (e.g. NSAIDs) and this can be remedied by a home visit from a pharmacist who has details of the patient's hospitalisation.

There are some limitations to our study. Firstly, the same pharmacist (who was not blinded to the allocation of patients) assessed the DRPs in the intervention and control group at 90 days, introducing possible bias towards the intervention group at identifying possible DRPs. However, this was a secondary endpoint to the study and this would have no effect on the primary outcome measure of readmission to hospital. Secondly, our compliance question may have overestimated actual compliance. We chose not to use a fully validated self-reported compliance questionnaire (e.g. Morisky scale³⁹) as this was not a primary endpoint of the study and it was thought that focusing on each patient's compliance in depth during the home visit may have been counterproductive to the overall intervention program. Thirdly, we may have underestimated the true

number of drug-related readmissions due to the retrospective assessment and by only including drug-related readmissions if they were clearly documented in the medical records. Fourthly, our study consisted of a relatively small sample that limited some of our interpretations and prevented us from determining exactly which patients may have benefited most from the home visit. Further research is required in larger randomised controlled trials to assess the targeting and cost-effectiveness of such intervention programs.

We believe that elderly patients with complex medical problems and prescribed multiple medications should routinely receive a home visit by a suitably qualified pharmacist who has the ability to liaise with the patient's general practitioner, community pharmacist and hospital following discharge. The pharmacist in this study was not an employee at the hospital and worked as a casual locum pharmacist, suggesting that it is not necessary to have extensive clinical skills but merely an understanding of what can go wrong after discharge and an ability to communicate effectively to all parties involved in the patient's care in the hospital and community.

While training and accreditation processes are important to ensure that pharmacists possess adequate clinical expertise when performing medication review, DRPs in the community are not complex and a perceived lack of clinical knowledge should not act as a barrier to community pharmacists becoming more involved in patient-oriented activities.²⁵ What is more important is that pharmacists show an interest and take the time to ask the right questions and liaise with caregivers to sort out relatively simple problems that could have hazardous consequences. Preconceived ideas that doctors will not welcome advice or suggestions were not borne out in this study. Most general practitioners appreciated the interest of another professional in the care of their patient.

CONCLUSION

This study has demonstrated the usefulness of a pharmacist in assisting with the management of medications after discharge from hospital. Pharmacist follow-up of high-risk elderly patients at home resulted in a reduction in DRPs and unplanned readmissions. The outcomes suggest that this sort of program merits more widespread application in Australian hospitals.

More assistance needs to be given to patients, particularly those with complex medical problems and prescribed multiple medications, when they are discharged from hospital.

A home-based pharmacy assessment can reveal many problems with medication management not otherwise detected easily.⁴⁰⁻⁴³ These assessments can lead to potentially useful interventions that can improve medication regimens and patient compliance. These programs may be particularly cost-effective if applied selectively to patients with a history of frequent unplanned hospital admission.⁴⁰

Ideally, the Home Medicines Review scheme could fund hospital pharmacists to perform, or assist in, medication reviews for high-risk patients soon after discharge from hospital. Community pharmacists would find it difficult to perform these types of review in isolation and without having access to comprehensive information on the patient's hospital admission.

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CONGRATULATIONS TO THE RECENT RECIPIENTS OF SHPA GRANTS

Grant	Recipient	Purpose	Value
BMS Clinical Pharmacy Research Grant	Michael Barras	To study individualised dosing of enoxaparin	\$8000
BMS Clinical Pharmacy Research Grant	Robert Marshall	To study population pharmacokinetics of risperidone in children	\$8000
DBL Development Fund	Stephen Haas	Attend European Society of Cardiology Congress, Vienna	\$2000
DBL Development Fund	Stephen Fowler	Attend introductory seminar in critical care, Melbourne	\$900
DBL Development Fund	Helen Leach	Attend 3rd ISOPP Australian regional meeting, Gold Coast	\$900
DBL Development Fund	Vicki Wilmott	Attend 3rd ISOPP Australian regional meeting, Gold Coast	\$900
MSD Pharmacy Specialisation Grants	Sue Scott	Study and sit for Certification in Geriatric Pharmacy examination	\$1500